

## SECTION - A - MCQs

1. b) 8
2. b) Contains all elements under consideration.
3. a)  $4/5$ .
4. c) Many-one relation
5. b)  $1/2$
6. a) 10
7. b)  $A \cap B = \emptyset$
8. c) Transitive

## SECTION - B - Very Short Answers

- 9.) A set is a collection of objects written in brackets.
- 10.) Domain of  $f(x) = x^2$  is real numbers and range is also real numbers.
- 11.)  $\sin 45^\circ \cos 45^\circ = 1/2$
- 12.)  $SP2 = 5 \times 4 = 20$ .
- 13.) Numbers of elements in power set is 16.
- 14.) One-one function is  $f(x) = x$  and many-one function is  $f(x) = x^2$ .

## Section C - Short Answer

15.  $LHS = \sin \theta / (1 + \cos \theta) + (1 + \cos \theta) / \sin \theta$ .
16.  $A \cup B = (1, 2, 3, 4, 5, 6)$  and  $A \cap B = (3, 4)$ .
17. Total numbers formed =  $5P3 = 60$ .
18. Relation is reflexive and symmetric.

## SECTION D - Long Answer.

19.  $(1 + \tan^2 \theta) (1 + \sin^2 \theta)$